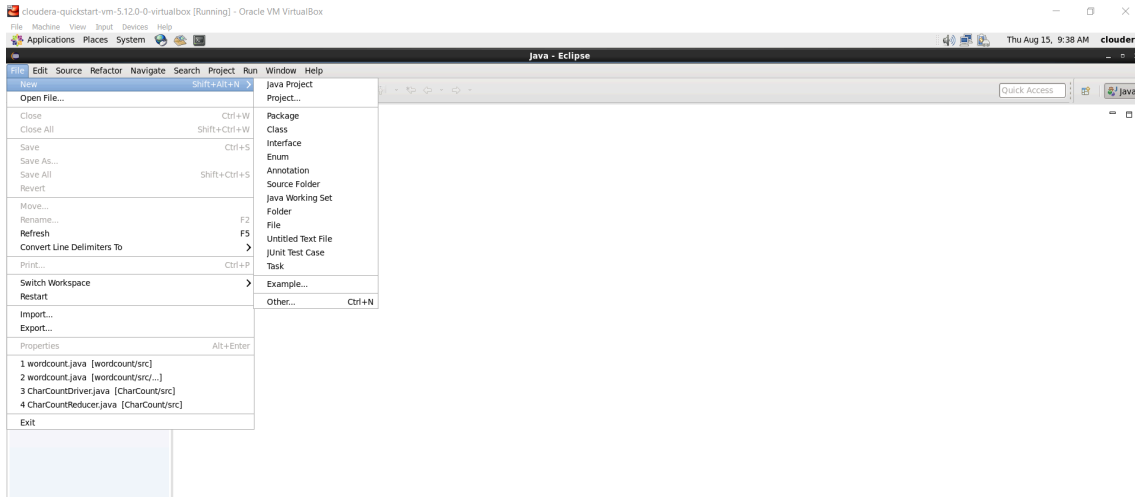
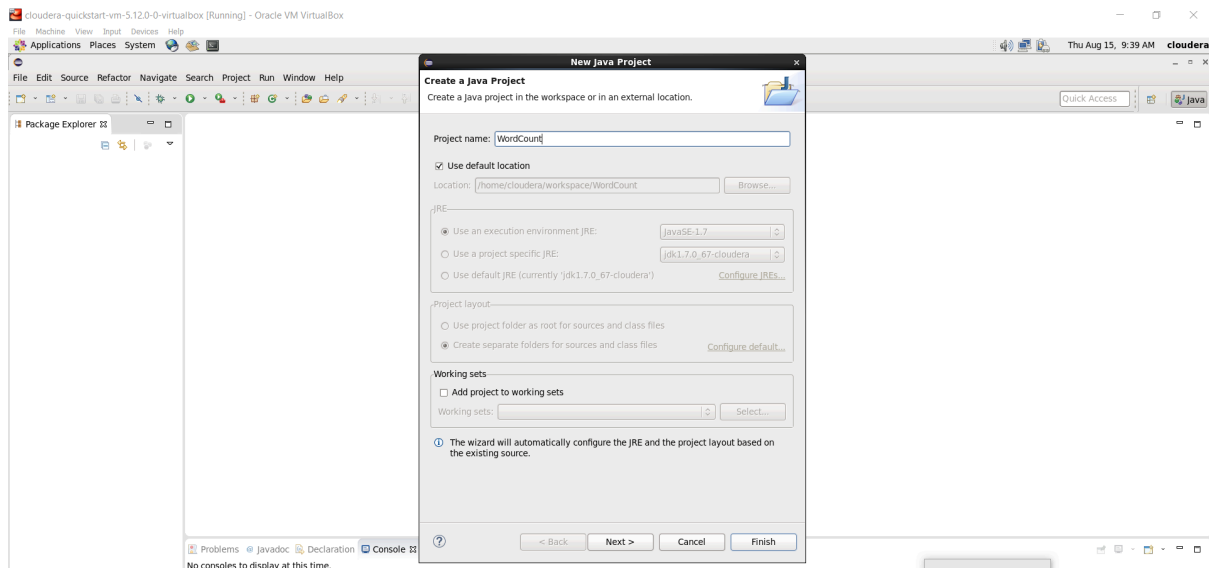


MAP REDUCE PROGRAMS – STEPS EXECUTION

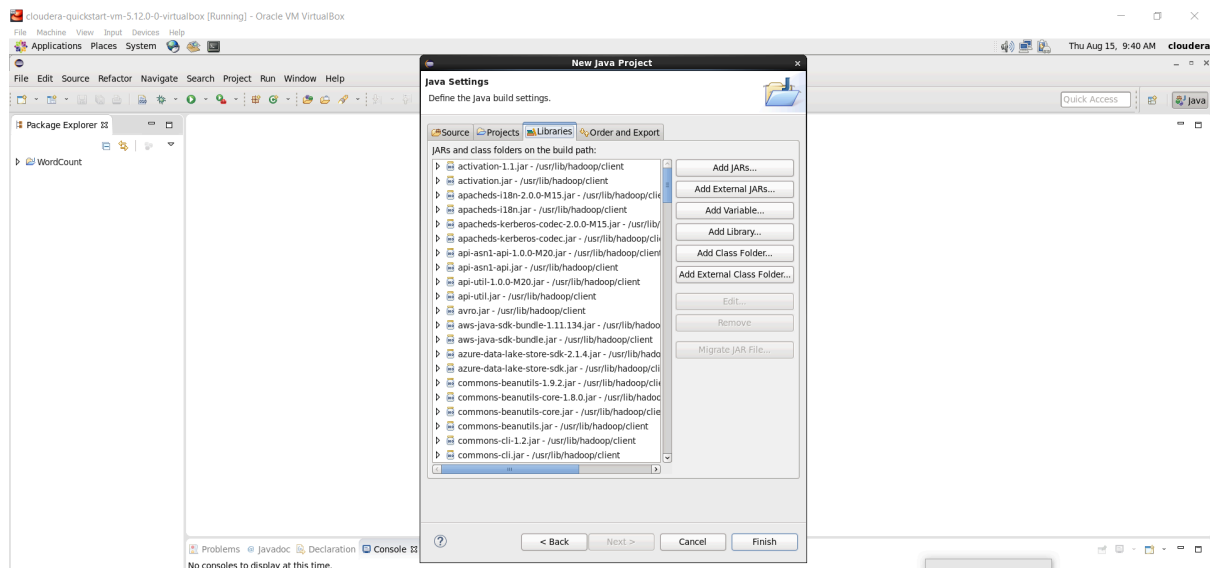
1. Open Eclipse and Click New Java Project



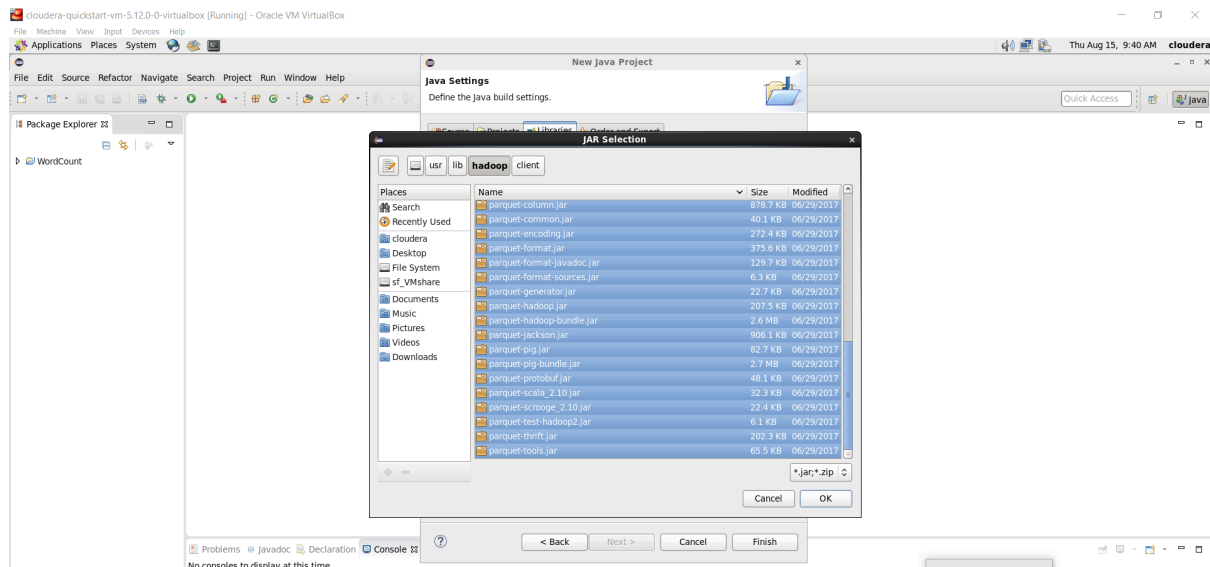
2. Enter Project Name – Eg : WordCount



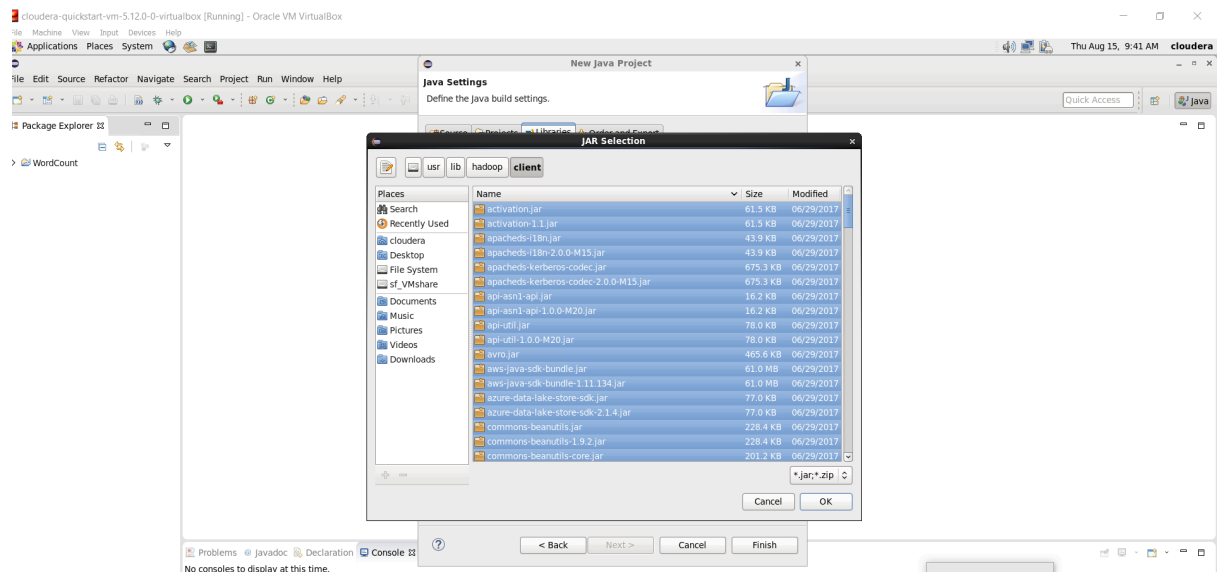
3. Click Libraries and click Add External JARS



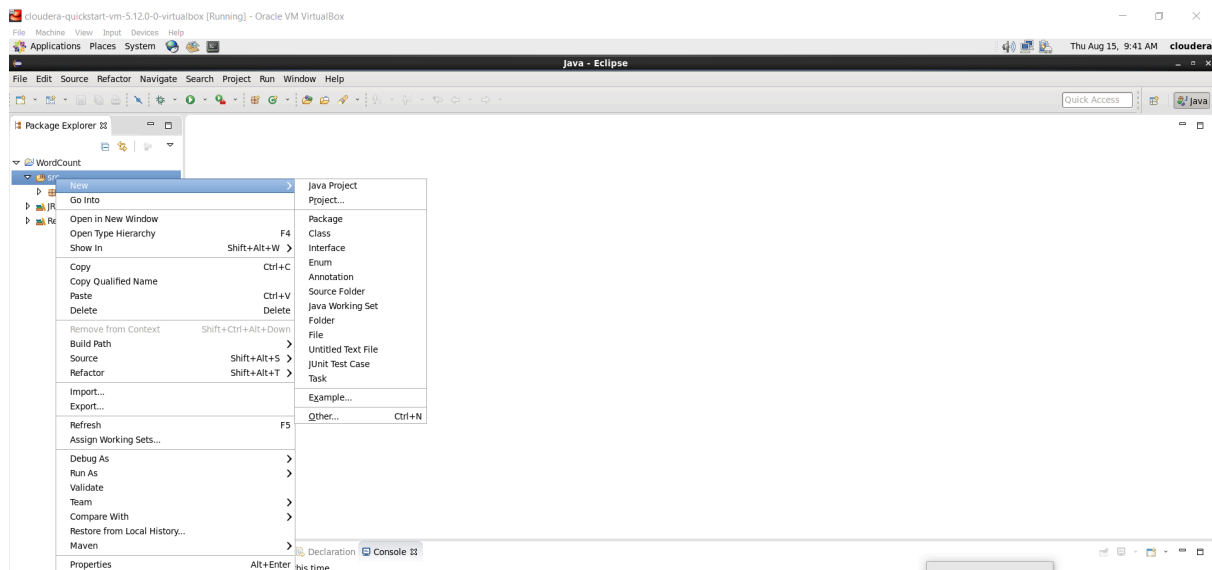
4. Select the file path – File System ☐ usr ☐ lib ☐ hadoop – Select all jar files and click OK.



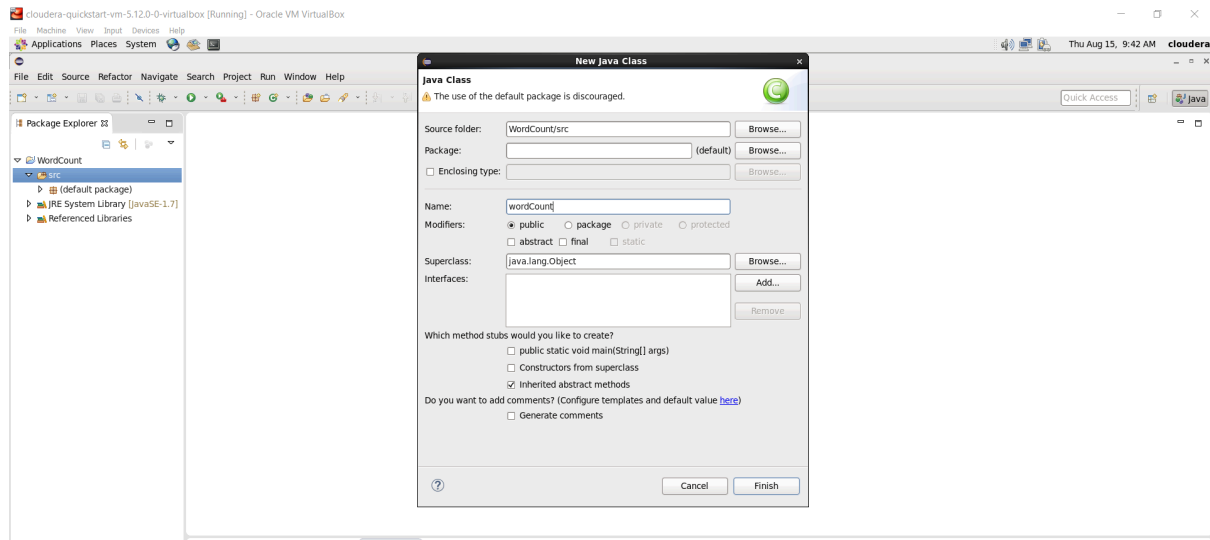
5. Again Click Libraries and click Add External JARS. Select the file path – File System
usr lib hadoop client – Select all jar files and click OK and click Finish.
Now, You can see Word Count Project in Package Explorer tab.



6. Right Click in src and select new class.



7. Type the java file name in Name filed – Eg : wordcount. Ensure that class name and java file name are same (case sensitive)

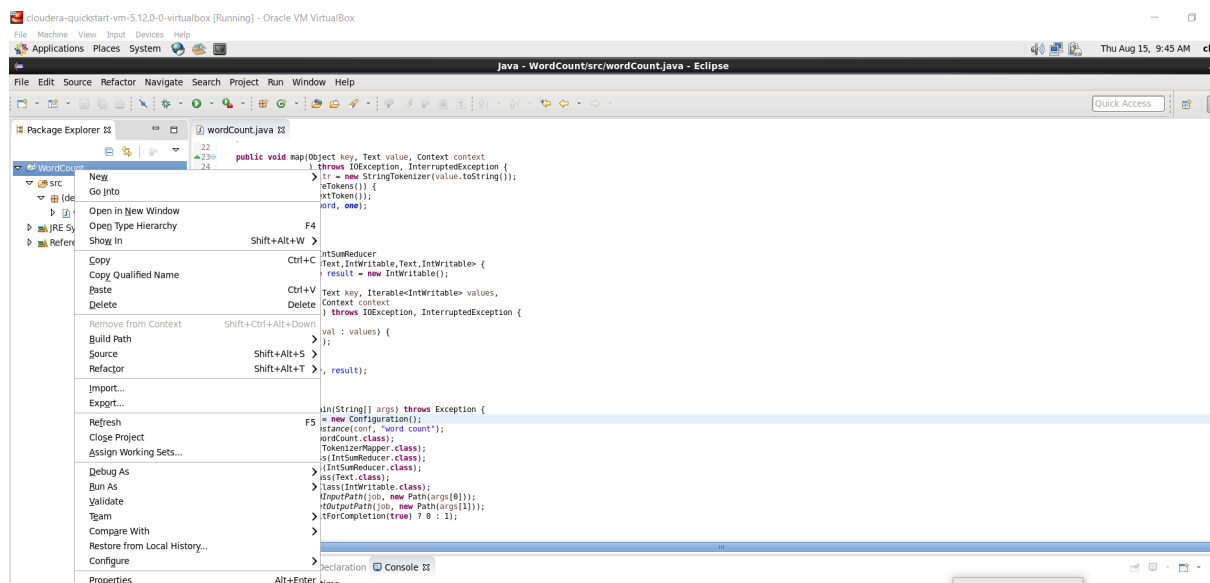


8. Now, Open the below mentioned link to copy the Word count MapReduce Java source code

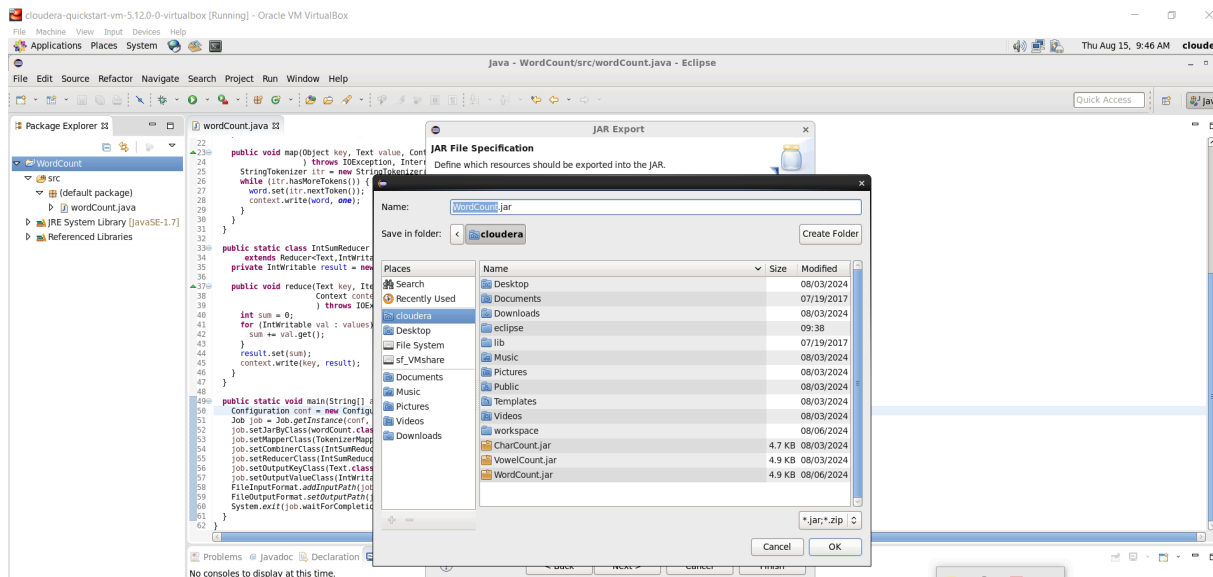
<https://hadoop.apache.org/docs/current/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduceTutorial.html>

Copy the source code and paste in wordcount.java file and click save. Ensure that there are no errors. If error, there is some mistake in creating project, delete and repeat the steps from first.

9. If no errors in project folder, right click on WordCount and click Export and select Java ☐ Jar file.



10. Click browse and Select the local file path to save the Jar file. Click cloudera and type the jar file name as WordCount.jar and click OK and click next and finish.



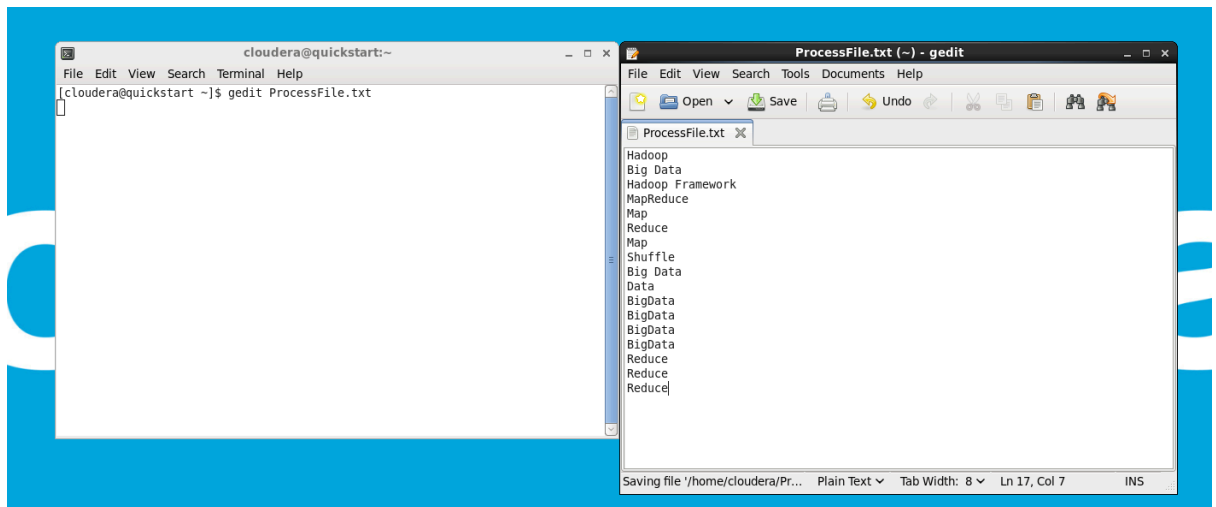
11. Now you can find the WordCount.jar file in the saved location.

12. Now open cloudera terminal and type the following commands to execute the jar file for the input text file and store the output in output folder.

- gedit ProcessFile.txt
Type some contents in the text file and click save.
- hdfs dfs -ls /
- hdfs dfs -mkdir /inputfolder
- hdfs dfs -ls /
- hdfs dfs -copyFromLocal /home/cloudera/ProcessFile.txt /inputfolder
- hdfs dfs -ls /inputfolder
- hadoop jar /home/cloudera/WordCount.jar wordCount /inputfolder/ProcessFile.txt /outputfolder
- hdfs dfs -ls /outputfolder
- hdfs dfs -cat /outputfolder/part-r-00000

13. In outputfolder, if two files are there Success and part-r-00000, then the code has been executed successfully. If any java error and exception check the java file name and class name both should be the same.

14. Terminal Output Screenshots:



```
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ gedit ProcessFile.txt
[cloudera@quickstart ~]$ hdfs dfs -ls /
bash: hdfs: command not found
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 9 items
drwxr-xr-x - cloudera supergroup      0 2024-08-08 03:14 /HiveDirectory
drwxrwxrwx - hdfs      supergroup      0 2017-07-19 05:34 /benchmarks
drwxr-xr-x - hbase     supergroup      0 2024-08-15 21:21 /hbase
drwxr-xr-x - cloudera supergroup      0 2024-08-07 23:57 /input
-rw-r--r-- 1 cloudera supergroup    40 2024-08-07 23:48 /input^C
drwxr-xr-x - solr      solr            0 2017-07-19 05:37 /solr
drwxrwxrwt - hdfs     supergroup      0 2024-08-03 00:54 /tmp
drwxr-xr-x - hdfs     supergroup      0 2017-07-19 05:36 /user
drwxr-xr-x - hdfs     supergroup      0 2017-07-19 05:36 /var
[cloudera@quickstart ~]$ hdfs dfs -mkdir /inputfolder
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 10 items
drwxr-xr-x - cloudera supergroup      0 2024-08-08 03:14 /HiveDirectory
drwxrwxrwx - hdfs     supergroup      0 2017-07-19 05:34 /benchmarks
drwxr-xr-x - hbase    supergroup      0 2024-08-15 21:21 /hbase
drwxr-xr-x - cloudera supergroup      0 2024-08-07 23:57 /input
-rw-r--r-- 1 cloudera supergroup    40 2024-08-07 23:48 /input^C
drwxr-xr-x - cloudera supergroup      0 2024-08-15 21:31 /inputfolder
drwxr-xr-x - solr     solr            0 2017-07-19 05:37 /solr
drwxrwxrwt - hdfs    supergroup      0 2024-08-03 00:54 /tmp
drwxr-xr-x - hdfs    supergroup      0 2017-07-19 05:36 /user
drwxr-xr-x - hdfs    supergroup      0 2017-07-19 05:36 /var
[cloudera@quickstart ~]$

[cloudera@quickstart ~]$ hdfs dfs -copyFromLocal /home/cloudera/ProcessFile.txt /inputfolder
[cloudera@quickstart ~]$ hdfs dfs -ls /inputfolder
Found 1 items
-rw-r--r-- 1 cloudera supergroup    133 2024-08-15 21:33 /inputfolder/ProcessFile.txt
[cloudera@quickstart ~]$
```

```

[cloudera@quickstart ~]$ hadoop jar /home/cloudera/WordCount.jar wordCount /inputfolder/ProcessFile.txt /outputfolder
24/08/15 21:34:35 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
24/08/15 21:34:36 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
24/08/15 21:34:36 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:952)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:690)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:879)
24/08/15 21:34:36 INFO input.FileInputFormat: Total input paths to process : 1
24/08/15 21:34:36 INFO mapreduce.JobSubmitter: number of splits:1
24/08/15 21:34:37 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:952)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:690)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:879)
24/08/15 21:34:37 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1723782016008_0001
24/08/15 21:34:37 INFO impl.YarnClientImpl: Submitted application application_1723782016008_0001
24/08/15 21:34:38 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1723782016008_0001/
24/08/15 21:34:38 INFO mapreduce.Job: Running job: job_1723782016008_0001
24/08/15 21:34:51 INFO mapreduce.Job: Job job_1723782016008_0001 running in uber mode : false
24/08/15 21:34:51 INFO mapreduce.Job: map 0% reduce 0%
24/08/15 21:34:58 INFO mapreduce.Job: map 100% reduce 0%
24/08/15 21:35:06 INFO mapreduce.Job: map 100% reduce 100%
24/08/15 21:35:07 INFO mapreduce.Job: Job job_1723782016008_0001 completed successfully
24/08/15 21:35:08 INFO mapreduce.Job: Counters: 49
    File System Counters
        FILE: Number of bytes read=123
        FILE: Number of bytes written=250253
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=257
        HDFS: Number of bytes written=81
        HDFS: Number of read operations=6
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=2
    Job Counters
        Launched map tasks=1
        Launched reduce tasks=1
        Data-local map tasks=1
        Total time spent by all maps in occupied slots (ms)=5297

```

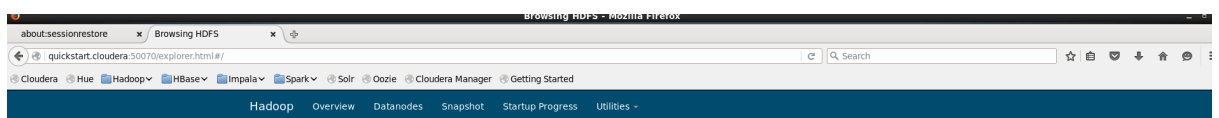


```

[cloudera@quickstart ~]$ hdfs dfs -ls /outputfolder
Found 2 items
-rw-r--r-- 1 cloudera supergroup 0 2024-08-15 21:35 /outputfolder/_SUCCESS
-rw-r--r-- 1 cloudera supergroup 81 2024-08-15 21:35 /outputfolder/part-r-000000
[cloudera@quickstart ~]$ hdfs dfs -cat /outputfolder/part-r-000000
Big 2
BigData 4
Data 3
Framework 1
Hadoop 2
Map 2
MapReduce 1
Reduce 4
Shuffle 1
[cloudera@quickstart ~]$ █

```

15. Cloudera browser Output folder screen shots



Browse Directory

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
drwxr-xr-x	cloudera	supergroup	0 B	Thu Aug 08 03:14:51 -0700 2024	0	0 B	HiveDirectory
drwxrwxrwx	hdfs	supergroup	0 B	Wed Jul 19 05:34:46 -0700 2017	0	0 B	benchmarks
drwxr-xr-x	hbase	supergroup	0 B	Thu Aug 15 21:21:30 -0700 2024	0	0 B	hbase
drwxr-xr-x	cloudera	supergroup	0 B	Wed Aug 07 23:57:18 -0700 2024	0	0 B	input
-rw-r--r--	cloudera	supergroup	40 B	Wed Aug 07 23:48:10 -0700 2024	1	128 MB	input^C
drwxr-xr-x	cloudera	supergroup	0 B	Thu Aug 15 21:33:34 -0700 2024	0	0 B	inputfolder
drwxr-xr-x	cloudera	supergroup	0 B	Thu Aug 15 21:35:05 -0700 2024	0	0 B	outputfolder
drwxr-xr-x	solr	supergroup	0 B	Wed Jul 19 05:37:04 -0700 2017	0	0 B	solr
drwxrwxrwt	hdfs	supergroup	0 B	Sat Aug 03 00:54:55 -0700 2024	0	0 B	tmp
drwxr-xr-x	hdfs	supergroup	0 B	Wed Jul 19 05:36:36 -0700 2017	0	0 B	user
drwxr-xr-x	hdfs	supergroup	0 B	Wed Jul 19 05:36:28 -0700 2017	0	0 B	var

about:sessionrestore x / Browsing HDFS

quickstart.cloudera:50070/explorer.html#/inputfolder

Cloudera Hue Hadoop HBase Impala Spark Solr Oozie Cloudera Manager Getting Started

Hadoop Overview Datanodes Snapshot Startup Progress Utilities

Browse Directory

/inputfolder Go!

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	cloudera	supergroup	133 B	Thu Aug 15 21:33:34 -0700 2024	1	128 MB	ProcessFile.txt

Hadoop, 2017.

Applications Places System

Browsing HDFS - Mozilla Firefox

about:sessionrestore x / Browsing HDFS

quickstart.cloudera:50070/explorer.html#/outputfolder

Cloudera Hue Hadoop HBase Impala Spark Solr Oozie Cloudera Manager Getting Started

Hadoop Overview Datanodes Snapshot Startup Progress Utilities

Browse Directory

/outputfolder Go!

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	cloudera	supergroup	0 B	Thu Aug 15 21:35:05 -0700 2024	1	128 MB	_SUCCESS
-rw-r--r--	cloudera	supergroup	81 B	Thu Aug 15 21:35:05 -0700 2024	1	128 MB	part-r-00000

Hadoop, 2017.

16. Repeat the steps to execute any Map Reduce Program in Hadoop framework.